

Chapter 1

General introduction and objectives

Transmission Electron Microscopy (TEM) has transformed materials science through the ability to directly view atomic structures. From Ruska and Knoll's original ideas [1], TEM has evolved from a mere imaging technique to a full analytical workstation. Through advances such as aberration correction [2], cold field emission sources [3], and ultra-stable monochromators [4], modern instruments are now capable of easily achieving sub-angstrom resolution and detecting picometer-scale displacements. Scanning Transmission Electron Microscopy (STEM), for instance, supports element-sensitive contrast utilizing detectors such as High-Angle Annular Dark Field (HAADF) and Annular Bright Field (ABF), providing Z (atomic number) - contrast imaging that is ideal for the identification of single atomic columns [5, 6]. Spectroscopic tools such as Electron Energy Loss Spectroscopy (EELS) allow us to examine materials' chemistry, and electronic properties. These are greatly valued when exploring functional oxides, semiconductors, and nanostructures in which the local behavior at the atomic level determines overall characteristics.

The growing interest in TEM studies over the past few decades is mainly driven by major technological advancements [7]. TEM is no longer restricted to vacuum static imaging but is extensively used today in dynamic in situ experiments, where researchers are able to perform experiments on materials under gas flow, liquid state, or external stimuli like temperature, biasing, or light [8]. Such advances have tremendously broadened the areas of TEM ap-

plication in nanotechnology, catalysis, battery studies, soft matter, and even in biological and medical sciences [9, 10]. In addition, combination with direct-electron detectors and advanced machine learning algorithms for data analysis, has improved resolution and speed, further expanding the limits of what TEM can achieve [11].

Despite these strengths, interpreting experimental images and spectra is a complex task. Image contrast depends sensitively on parameters like sample thickness, defocus, aberrations, and dynamical scattering, which complicate direct analysis. To address this, simulations serve as an essential interpretation tool. For example, the multislice method of STEM image simulation, by considering the wave propagation of electrons through a crystal model divided into slices, provides realistic images that can be compared with experimental images [12]. These simulations help confirm structural models by predicting how a given arrangement should appear under specific imaging conditions. Likewise, Density Functional Theory (DFT) has long been a standard for simulating electronic structure and optical response. DFT simulations provide insights about the electronic structure, material properties, and chemical processes, supporting both fundamental research and technological development. It also enables simulation of EELS, including fine structures such as Electron Energy Loss Near Edge Structure (ELNES), allowing researchers to distinguish oxidation states and chemical environments [13, 14]. Meanwhile, image analysis tools such as ATOMAP [15] enable the determination of atomic column positions and shapes in STEM images, even when the images contain many columns composed of a variety of atomic species with large intensity differences. Together, these tools provide a synergistic framework, where experimental observations are supported by simulation and quantitative analysis, enabling a deeper understanding of both crystal structures and electronic interactions.

This thesis is driven by the motivation of advancing the interpretation of transmission electron microscopy data through the simulation of STEM images and DFT (EELS) spectra. By focusing on how atomic configurations and bonding environments influence contrast in TEM/STEM and spectral features in EELS, the work contributes to a deeper understanding of the data obtained through electron microscopy. The approach is applied across a variety of materials of high technological interest.

The objectives of the thesis are as follows:

1. Introduce and establish the theoretical and computational frameworks for STEM imaging and EELS simulations.

By explaining the main theoretical concepts and computational approaches used in the thesis, we aim to establish the foundation for the analyses carried out to solve specific materials science problems. We will describe the process of image simulation using multislice techniques, present STEM image analysis techniques using ATOMAP, and describe the theoretical underpinnings of DFT and how it is used to simulate EELS spectra and ELNES features.

2. Investigate the influence of oxygen stoichiometry and structural phase transitions on the dielectric properties of hafnia-based resistive memory devices.

The objective is to comprehend the structural alterations brought about by oxidation in hafnium oxide compounds, such as phases transition from rhombohedral to monoclinic, and its impact on the dielectric characteristics of Resistive Random Access Memory (RRAM) devices through the comparison of DFT simulations with experimental STEM and EELS data. This knowledge will help link local atomic structural transformations with dielectric performance, contributing to the design of next-generation memory materials.

3. Examine the effects of pressure and charge on the EELS spectral features of xenon gas clusters.

The aim is to evaluate the effects of changes in external parameters, namely electronic charge and pressure, on the electronic structure and associated spectral signatures in confined Xe₂ and Xe₅ clusters. Changes in the M_{4,5} edge will be interpreted using DFT-based EELS simulations. These insights are essential for the interpretation of EELS spectra in systems where small changes in environment can lead to measurable spectral shifts.

4. Develop a methodology to quantify antisite defects in $\text{La}_{0.8}\text{Sr}_{0.2}\text{MnO}_3$ (LSMO) using STEM image simulations and intensity analysis.

This goal is to quantify antisite defect densities in LSMO using a Relative Intensity Ratio (RIR) metric by combining multislice simulations, ATOMAP analysis, and HAADF-STEM imaging. In order to comprehend trends in defect formation, this technique will be used on LSMO thin films that have been synthesised under various oxygen partial pressures. Understanding defect distributions as a function of synthesis conditions, such as oxygen partial pressure, can help optimise the performance of LSMO in solid state electrochemical devices.

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